

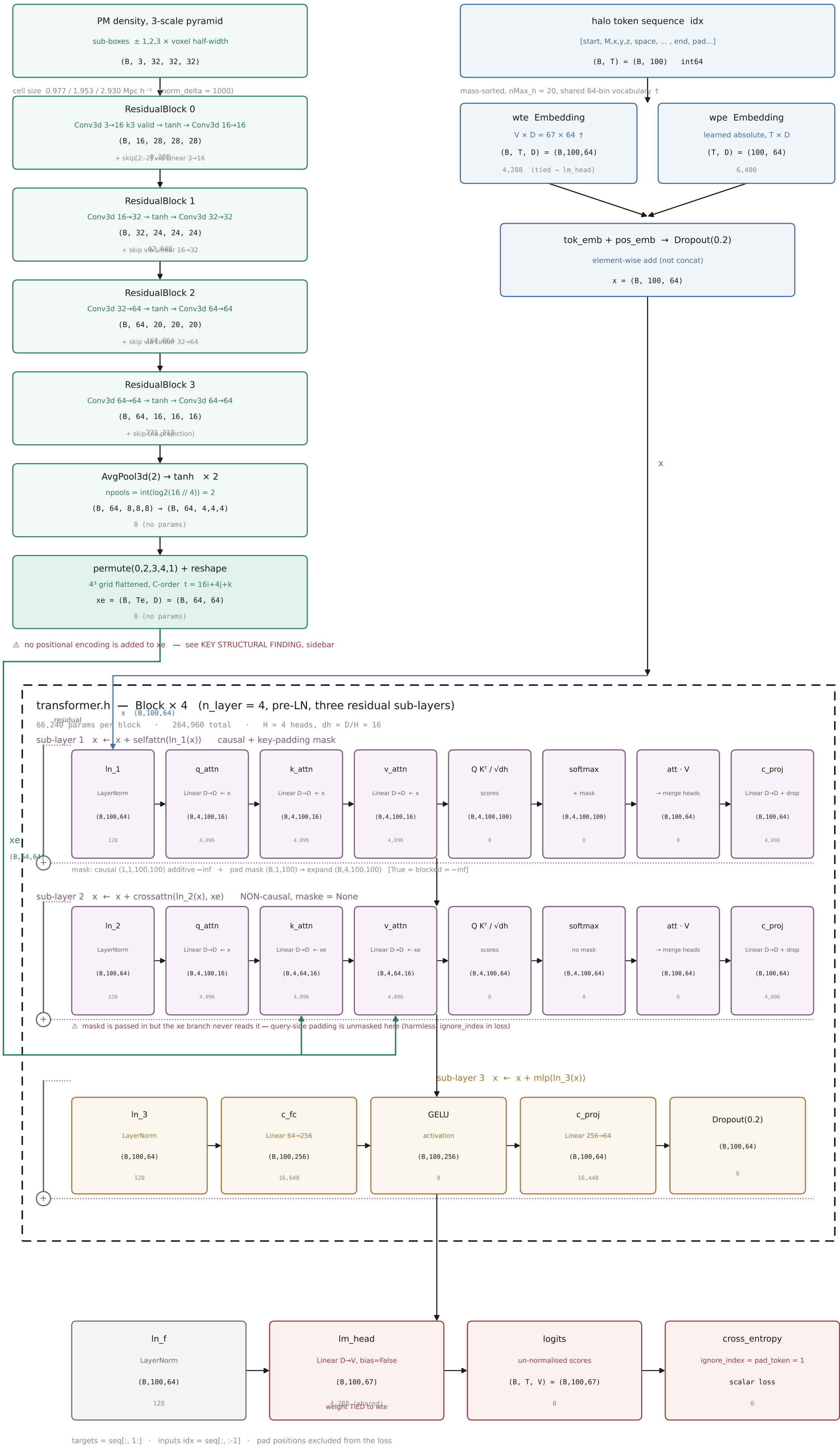
GOTHAM — training forward pass

Generative cOnditional Transformer for Halo's Auto-regressive Modeling · repo shivampcosmo/GOTHAM @ 7ef09cc · entry point src/train.py

All tensor shapes VERIFIED by forward hooks (torch 2.2.2, CPU, B=2). Inferred values are marked † and listed bottom-right.

CONDITIONING PATH

SEQUENCE PATH



SHAPE LEGEND

B	batch (train.py: 1950)
T	100 decoder seq len = max_sentence_length - 1 †
Te	64 encoder memory tokens = density_grid_out³ = 4³
D	64 n_embd
H	4 n_head
dh	16 = D / H
V	67 vocab = nvocab + 3 †
C	3 density channels
G	32 density_grid_in

PARAMETERS (hook-verified)

cnn3D	439,712
ResidualBlock 0	8,288
ResidualBlock 1	42,048
ResidualBlock 2	168,064
ResidualBlock 3	221,312
transformer.wte	4,288
transformer.wpe	6,400
transformer.h (x4)	264,960
per Block	66,240
selfattn	16,384
crossattn	16,384
mlp	33,088
3 × LayerNorm	384
transformer.ln_f	128
lm_head	(tied)
TOTAL	715,488
encoder share	61.5 %
dtype	cnn bf16 / rest fp32

CONFIG (src/train.py)

n_layer	4
n_head	4
n_embd	64
dropout	0.2
bias	True
ksize	3
density_grid_in	32
density_grid_out	4
ninp_density	3
flash	False
pad_token	1 (from h5)
batch_size	1950
learning_rate	2e-3 cosine
optimizer	AdamW

† INFERRED, NOT READ

T = 100 and V = 67 come from prep_data_FPM.ipynb (nMax_h = 20, nvocab = 64). train.py reads both from an .h5 that is NOT in the repo, so they could differ at run time.

Everything else was read from code or measured by hooks.

CONFIG CONFLICTS

The 4 checkpoints in model_checkpoints/ do NOT match src/train.py: they have n_embd=48, vocab=131 (nvocab=128), block_size=161, n_layer=4 — i.e. src/arxiv/train.py (nMax_h = 32). best_val_loss 2.7638 @ iter 3920.

Every run_scripts/*.sh invokes src/test_ddp*.py, but those files now live in src/arxiv/; and submit_run_PM_Mmin_13p5.sh calls a file absent from the repo entirely.

No launcher references src/train.py. It is used here as the newest, most complete trainer — a judgement call.

KEY STRUCTURAL FINDING

No positional encoding is added to the 64 memory tokens xe. Cross-attention is a weighted sum over keys, so permuting xe leaves the output bit-identical. Measured: max|logits(xe) - logits(xe[perm])| = 3.0e-07 (fp32 epsilon)

The encoder is translation-equivariant (valid padding, no boundary cue), so the decoder receives an unordered multiset of local density features, not a map.

Encoder geometry: RF 20 cells = 19.5 Mpc/h, stride 4 = 3.9 Mpc/h; the 4³ token centres span only the central 37% of the voxel.

TRACE PROVENANCE

torch 2.2.2 CPU, B = 2, seed 0. Encoder cast to fp32 for the trace: bf16 avg_pool3d is CUDA-only. Shapes are dtype-invariant. xformers import stubbed (unused).

No repo plotting palette exists; colours chosen here by role.